AI Assisted Coding Assignment

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Lab Assignment 15.3

Task 1: Online Food Ordering API

Question: Design a REST API for an online food ordering system with CRUD operations.

Prompt:

"Generate a Python Flask REST API for an online food ordering system with endpoints for menu listing, placing orders, updating, tracking, and deleting orders. Include comments and basic validation."

Code:

```
from flask import Flask, jsonify, request

app = Flask(__name__)

menu = [

    {"id": 1, "name": "Pizza", "price": 250},

    {"id": 2, "name": "Burger", "price": 120}

]

orders = []

@app.route('/menu', methods=['GET'])

def get_menu():

    return jsonify(menu)

@app.route('/order', methods=['POST'])

def place_order():

    data = request.json

    order = {"id": len(orders)+1, "items": data["items"], "status": "Placed"}

    orders.append(order)

    return jsonify(order)

@app.route('/order/<int:id>', methods=['GET'])

def get_order(id):
```

```
    for order in orders:
        if order["id"] == id:
            return jsonify(order)
    return jsonify({"error": "Order not found"}), 404
@app.route('/order/<int:id>', methods=['PUT'])
def update_order(id):
    data = request.json
    for order in orders:
        if order["id"] == id:
            order["items"] = data["items"]
            return jsonify(order)
    return jsonify({"error": "Order not found"}), 404
@app.route('/order/<int:id>', methods=['DELETE'])
def delete_order(id):
    for order in orders:
        if order["id"] == id:
            orders.remove(order)
            return jsonify({"message": "Order deleted"})
    return jsonify({"error": "Order not found"}), 404
app.run(debug=True)
```

Output:

GET /menu →

```
[
 {"id":1,"name":"Pizza","price":250},
 {"id":2,"name":"Burger","price":120}
]
```

POST /order →

```
{"id":1,"items":["Pizza"],"status":"Placed"}
```

Code Explanation:

* Flask is used to create REST API.

* `/menu` returns list of dishes.

* `/order` creates a new order.

* `/order/<id>` supports GET, PUT, DELETE.

* Data stored in in-memory lists.

Task 2: Employee Payroll API

Question: Create an API to manage employees and salaries.

Prompt:

"Generate a Flask API for employee payroll management with endpoints for adding employees, updating salary, deleting employees, and listing all employees."

Code:

```
from flask import Flask, jsonify, request

app = Flask(__name__)

employees = []

@app.route('/employees', methods=['GET'])

def get_employees():

    return jsonify(employees)

@app.route('/employees', methods=['POST'])

def add_employee():

    data = request.json

    emp = {

        "id": len(employees)+1,

        "name": data["name"],

        "salary": data["salary"]

    }

    employees.append(emp)

    return jsonify(emp)

@app.route('/employees/<int:id>/salary', methods=['PUT'])

def update_salary(id):

    data = request.json

    for emp in employees:

        if emp["id"] == id:
```

```
        emp["salary"] = data["salary"]
        return jsonify(emp)
    return jsonify({"error": "Employee not found"}), 404
@app.route('/employees/<int:id>', methods=['DELETE'])
def delete_employee(id):
    for emp in employees:
        if emp["id"] == id:
            employees.remove(emp)
            return jsonify({"message": "Employee deleted"})
    return jsonify({"error": "Employee not found"}), 404
app.run(debug=True)
```

Output:

POST /employees →

{"id":1,"name":"John","salary":50000}

PUT /employees/1/salary →

{"id":1,"name":"John","salary":60000}

Code Explanation:

* Stores employee data in list.
* Supports CRUD operations.
* Salary update handled separately.
* Includes validation for missing employee.

Task 3: Student Information REST API

Question: Build a Student Management API.

Prompt:

"Create a Flask REST API for student management with add, get, update, and delete operations."

Code:

```
from flask import Flask, jsonify, request
app = Flask(__name__)
students = []
@app.route('/students', methods=['POST'])
```

```
def add_student():
    data = request.json
    student = {
        "id": len(students)+1,
        "name": data["name"],
        "course": data["course"]
    }
    students.append(student)
    return jsonify(student)
@app.route('/students', methods=['GET'])
def get_students():
    return jsonify(students)
@app.route('/students/<int:id>', methods=['PUT'])
def update_student(id):
    data = request.json
    for s in students:
        if s["id"] == id:
            s["name"] = data["name"]
            return jsonify(s)
    return jsonify({"error": "Not found"}), 404
@app.route('/students/<int:id>', methods=['DELETE'])
def delete_student(id):
    for s in students:
        if s["id"] == id:
            students.remove(s)
            return jsonify({"message": "Deleted"})
    return jsonify({"error": "Not found"}), 404
app.run(debug=True)
```

Output:

POST /students →

{"id":1,"name":"Alice","course":"CSE"}

GET /students →

[{"id":1,"name":"Alice","course":"CSE"}]

Code Explanation:

* Manages student records.

* Uses list as database.

* CRUD operations implemented.

* Each student has unique ID.

Task 4: Weather Information API

Question: Integrate external weather API.

Prompt:

"Generate Flask API code to fetch current weather using OpenWeatherMap API."

Code:

```
import requests

from flask import Flask, jsonify

app = Flask(__name__)

API_KEY = "sk-proj-
vMZZoXTmGybZEzckB0RBCrBffI374lYS1gXN9cBuc3ZVd649SBCoRzx_D0i8AcXlB4EOmTs3V6T3BlbkFJffu
tICXnL37pRSaAdmc6tWO9OHQWusNZCcnzOhsgaybRnLb9SdQiaY5HJNJUmjfiisnOk_T44A"

@app.route('/weather/current/<city>', methods=['GET'])

def get_weather(city):

    url = f"http://api.openweathermap.org/data/2.5/weather?q={city}&appid={sk-proj-
vMZZoXTmGybZEzckB0RBCrBffI374lYS1gXN9cBuc3ZVd649SBCoRzx_D0i8AcXlB4EOmTs3V6T3BlbkFJffu
tICXnL37pRSaAdmc6tWO9OHQWusNZCcnzOhsgaybRnLb9SdQiaY5HJNJUmjfiisnOk_T44A}"

    response = requests.get(url).json()

    return jsonify({

        "temperature": response["main"]["temp"],

        "humidity": response["main"]["humidity"],

        "condition": response["weather"][0]["description"]

    })

app.run(debug=True)
```

Output:

GET /weather/current/Chennai →

```
{
 "temperature": 300,
 "humidity": 70,
 "condition": "clear sky"
}
```

Code Explanation:

* Uses requests library to call API.

* Extracts temperature, humidity, condition.

* Returns structured JSON response.

Task 5: Delete Item AP

Question: Create DELETE endpoint to remove item by index.

Prompt:

"Write Flask DELETE API to remove item from list using index with validation."

Code

```
from flask import Flask, jsonify
app = Flask(__name__)
items = [{"name": "Notebook", "price": 250}]
@app.route('/items/<int:index>', methods=['DELETE'])
def delete_item(index):
    if index < 0 or index >= len(items):
        return jsonify({"error": "Item not found"}), 404
    removed_item = items.pop(index)
    return jsonify({"message": "Item deleted", "item": removed_item})
@app.route('/items', methods=['GET'])
def get_items():
    return jsonify(items
app.run(debug=True)
```

Output:

DELETE /items/0 →

{"message": "Item deleted", "item": {"name":"Notebook","price":250}

GET /items →

[]

Code Explanatio

* Validates index range.

* Removes item using pop().

* Returns deleted item.

* Ensures error handling